

Examensarbete inom teknik

Grundnivå, 15 hp

Conceptual design of an eVTOL

A study of flight mechanics and economics with implementation in Singapore

Teodor Salwén

Rikard Holmberg Wilkens

1 Abstract

This study investigates the feasibility of implementing an eVTOL air taxi service in Singapore by analyzing flight mechanics, energy consumption, noise pollution and profitability possibilities. A quadcopter configuration was developed and evaluated using momentum theory as well as blade element momentum theory. Results show that cruise speeds between 140–170 km/h and mission energy requirements of approximately 75 kWh are achievable. Noise constraints limit operations, with daytime flights feasible at around 600 m altitude, while nighttime operations are largely impractical at this time.

The economic analysis shows positive unit economics under applied assumptions, with a profit per flight of approximately 108 USD. However, the high initial investment and resulting payback period of around 20 years indicate that profitability will only be possible in a long-term scenario.

The study concludes that while the system is technically feasible, economic viability depends on scaling, demand growth, and external support.

Contents

1 Abstract **2**

2 Background **5**

3 Symbols and Nomenclature **6**

4 Design **7**

5 Flight Mechanics **9**
5.1 Equilibrium and Performance 9
5.2 Momentum Theory 10
5.3 Blade Element Momentum Theory 11

6 Energy Consumption **14**
6.1 Motor 14
6.2 Battery analysis 14

7 Noise pollution **16**
7.0.1 Rotor tip speed and tip Mach number 16
7.0.2 Acoustic source strength 16
7.0.3 Rotor geometry contribution 17
7.0.4 Propagation distance correction 17
7.0.5 Directional radiation effects 17
7.0.6 Multiple rotor correction 17
7.0.7 Total sound pressure level 17
7.0.8 Perceived noise level 17
7.1 eVTOL in urban environment 18

8 Regulations **19**

9 Economic possibilities **20**
9.1 Background 20
9.2 Global outlook 20
9.3 Demand and market adoption in Singapore 21
9.4 Cost structure and unit economics 22
9.5 Achieve Profitability 23
9.5.1 Economic Analysis 23

10 Result **23**
10.1 Momentum Theory and BEMT 23
10.2 Noise Pollution 24
10.3 Economic Analysis 24

11 Discussion **24**
11.1 Noise Pollution 25
11.2 Economic Outlook 26

12 Conclusion **27**

13 Appendix **28**

A Open VSP Cross sectional area **28**

B Momentum Theory Assumptions **29**

C Drone model for approximation of Cd **30**

D Email from T-Power **31**

E Economic model equations **32**
E.1 Fleet utilization 32
E.2 Variable costs 32
E.3 Fixed costs 32
E.4 Total cost metrics 33
E.5 Revenue and profit 33
E.6 Payback time 33

F Use of Artificial Intelligence Tools **34**

2 Background

Urbanization and increasing population density are placing growing demands on transportation systems in major metropolitan regions. In dense urban environments, the expansion of road- and rail-based infrastructure is often constrained by limited available land, high construction costs, and long implementation timelines. As a result, congestion and travel time variability remain significant challenges in many large cities.

Airborne transportation offers a potential means of avoiding ground-based networks by utilizing the vertical dimension. However, conventional helicopters, while capable of vertical take-off and landing, are associated with high noise levels, substantial local emissions, and high operational and maintenance costs. These limitations have restricted their widespread adoption for regular passenger transport in urban settings.

In recent years, electric Vertical Take-Off and Landing (eVTOL) aircraft have been proposed as a key enabling technology for Urban Air Mobility. By employing electric propulsion and simplified mechanical solutions, eVTOL concepts aim to reduce noise emissions, eliminate exhaust emissions, and lower operating costs compared to conventional rotorcraft. In addition, many proposed eVTOL systems are designed for a high degree of automation, with the long term objective of fully autonomous operation. Ehang is one of many companies with this as the final goal [1]. Autonomous flight has the potential to further reduce operational costs, increase system scalability, and mitigate pilot availability constraints, which are significant factors for large scale air taxi deployment.

Several cities have begun exploring the implementation of eVTOL-based air taxi services. In particular, Dubai has positioned itself as an early adopter with goals of implementing the first flying taxi this year [2].

The purpose of this report is to study whether it is possible to implement an eVTOL system in Singapore. The main areas considered are structural design, flight mechanics, energy consumption, noise, and economic factors. These are the key aspects that need to be analysed to understand if eVTOL operations can be successfully deployed.

3 Symbols and Nomenclature

Symbol	Definition
r	Radial distance from the hub to a blade element
R, R_{hub}	Rotor radius and hub cut-out radius
V	Airspeed magnitude relative to the air
a_f	Rotor disk incidence angle (angle between the rotor axis and the freestream)
ψ	Azimuth angle of the blade element
U_t	Velocity component tangential to the plane of no-feathering (chordwise component)
U_p	Velocity component perpendicular to the plane of no-feathering (inflow component)
V_E	Effective resultant velocity at the blade element
v_{i0}	Disk-averaged induced velocity
Ω	Angular velocity of the rotor
$\beta(r)$	Local pitch angle of the blade element
ϕ	Inflow angle at the blade element
$\alpha(r, \psi)$	Local aerodynamic angle of attack of the blade element
α_{ZL}	Zero-lift angle of the airfoil section
N_b	Number of blades
$c(r)$	Local chord
V_{tip}	Rotor blade tip speed
V_∞	Freestream velocity
RPM	Rotational speed of the rotor (revolutions per minute)
D_p	Propeller diameter
M_{tip}	Tip Mach number
a	Speed of sound
γ	Ratio of specific heats for air
R	Gas constant
T	Ambient temperature
P_{HP}	Shaft power delivered to the propeller (horsepower)
N_B	Number of blades
x_{ft}	Distance between observer and aircraft (feet)
θ	Angle between rotor axis and observer
FL_1	Acoustic source strength contribution
FL_2	Rotor geometry contribution
FL_3	Propagation distance correction
DI	Directivity correction
NC	Multiple rotor correction
SPL	Total sound pressure level
PNL	Perceived noise level
PNL_A	A-weighted perceived noise level

Table 1: Definition of variables used in the BEMT formulation as well as in the Noise pollution model

Parameter	Value	Unit
Mission range	30	km
Energy per flight	75.6	kWh
Passengers per flight	2	-
Load factor	0.8	-
Fleet	10	eVTOLs
Flights annually	2000	flights/year per eVTOL
Electricity price	0.30	USD/kWh
Battery capacity	98.3	kWh
Battery cost	200	USD/kWh
Battery cycle life	2000	cycles
eVTOL purchase cost	200,000	USD
eVTOL lifetime	20,000	hours
Design and prototyping	20,000,000	USD
Infrastructure cost	100,000,000	USD
Certification cost	2,000,000	USD
Government subsidy	80%	of infrastructure
Maintenance cost	80	USD/flight
Ground staff cost	40	USD/flight
Insurance cost	30	USD/flight
Ticket price	250	USD/passenger
Ad/partnership revenue	30	USD/flight
Corporate Tax	17%	-

Table 2: Key assumptions used in the economic analysis

4 Design

The design of an eVTOL aircraft involves a highly multidimensional problem, where aerodynamic performance, propulsion efficiency, structural considerations, and energy storage must be balanced. To make the design process manageable, a set of primary design constraints and mission requirements are defined.

The primary objective is to develop an aircraft capable of transporting two passengers, each with an assumed mass of 80 kg, over a range of 30 km. This range is reasonable given the Singapore operating environment, as it allows the eVTOL to cover the city center as well as routes to Batam, Indonesia, which is a very lucrative route because of the high demand and the otherwise long ferry travel times. The estimated total take-off mass, including passengers, structure, propulsion system, and energy storage, is set to approximately 650 kg. Similar helicopters with comparable payload capacity typically weigh between 600 and 800 kg, which provides a reasonable basis for this estimate. [3] The design should be of a quadcopter style, with four symmetrical motors mounted along the horizontal line of the aircraft.

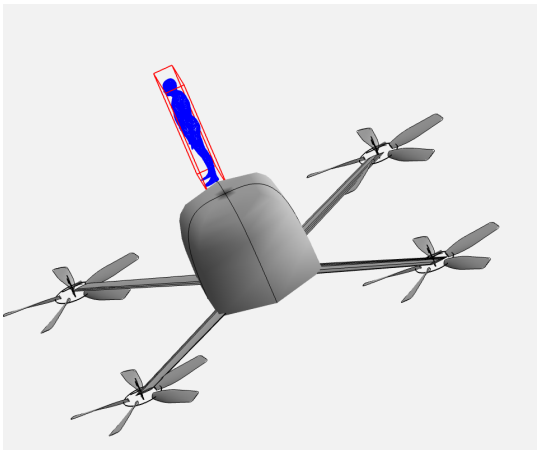
In the early stages of the project, NASA's design tool OpenVSP was used to obtain an initial estimate of the eVTOL configuration. [4] One advantage of OpenVSP is its ability to analyze flow characteristics and moments of inertia. However, the tool is primarily suited for fixed-wing aircraft and lacks support for structural elements such as beams, which are essential for eVTOL designs. Consequently, the only parameter retained from this model was the cross-sectional area at different pitch angles.

After the initial designs in OpenVSP, a more dedicated design was sketched using CAD software NX Design Center. The illustration of Figure 1c shows the resulting eVTOL concept,

where basic aerodynamic considerations were implemented. In particular, the main chassis was designed with a rounded shape to reduce drag. Early design considerations also included the use of two blades per rotor.

The next iteration of the eVTOL incorporated additional aerodynamic improvements. In particular, circular shrouds were added around the rotors to reduce tip losses, noise levels and to provide a divider for the humans near the aircraft. [5] The design also features a glass panel integrated into the main body, along with more refined and detailed rotor blades. These modifications contribute to improved aerodynamic efficiency while also enhancing the overall structural and visual design of the aircraft.

In the final iteration, the landing gear design was revised, transitioning from an initially unstable post-based configuration to a more robust helicopter-style landing gear. Apart from this modification, the design remains unchanged from the previous iteration.



(a) Initial eVTOL illustration from OpenVSP



(b) First CAD iteration from NX DesignCenter



(c) Improved CAD iteration with rotor shrouds



(d) Final CAD iteration with revised landing gear

Figure 1: Development process of the eVTOL concept, from the initial OpenVSP model to successive CAD iterations in NX DesignCenter.

5 Flight Mechanics

5.1 Equilibrium and Performance

The flight mechanics constitute the core functional aspect of the eVTOL aircraft. The layout of the eVTOL consists of four fixed motors symmetrically positioned around the fuselage, where the payload is seated. The motors generate a combined thrust force exceeding the gravitational force, enabling vertical take-off and hover.

During hovering flight, a forward pitch motion is achieved by increasing the thrust generated by the rear motors relative to the front motors, creating a torque about the body-fixed y-axis. This rotation defines the pitch angle θ, which is measured relative to the horizon. As the pitch angle increases, the vertical component of the thrust is reduced, which, if not compensated for, would result in a loss of altitude. To maintain altitude, the total thrust magnitude must therefore be increased such that the vertical force component balances the gravitational force. Increasing the thrust also increases the horizontal force component, which accelerates the vehicle forward and enables cruise flight. For steady hover, the rotors must be arranged in a counter-rotating configuration to cancel the net reaction torque generated by the propulsion system. In this configuration, the front-left and rear-right rotors rotate in the same direction, while the remaining pair rotates oppositely, resulting in a near-zero net yaw moment. Due to the symmetric layout of the aircraft, this configuration ensures stable hover without inducing rotational motion.

An alternative solution is the use of coaxial rotors, where two rotors are mounted on the same axis and rotate in opposite directions. This configuration eliminates the reaction torque locally at each propulsion unit, seen on EHangs EH216-s [6].

The thrust force components are given by

$$F_v = T\cos(\theta) = mg \qquad F_h = T\sin(\theta), \tag{1}$$

where T is the total thrust generated by the motors, as shown in Figure 2 below.

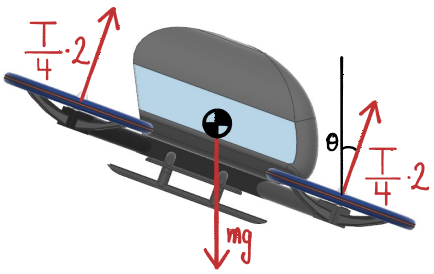


Figure 2: Free body diagram of eVTOL

From the vertical force balance, the required thrust magnitude is

$$T = \frac{mg}{\cos(\theta)}. \tag{2}$$

Consequently, the horizontal force component becomes

$$F_h = mg\tan(\theta). \tag{3}$$

The cruise speed depends on the pitch angle and is determined by the balance between the horizontal thrust component and the aerodynamic drag force. The drag force is modelled as

$$F_d = \frac{1}{2}\rho S C_d v_{\text{cruise}}^2, \tag{4}$$

where ρ is the air density, S is the projected area normal to the flow direction, C_d is the drag coefficient, and v_{cruise} is the cruise velocity [7].

The drag coefficients C_d are approximated from Hammer *et al.* [8], using the drone configuration alternative UAS 1, see appendix C. The drag coefficients are assumed to remain constant under geometric scaling. The resulting values of S are collected from the Open VSP illustration A. The drag coefficient as functions of pitch angle are presented in Table 3.

Pitch Angle (deg)	**Projected Area,** S **(m^2)**	**Drag Coefficient,** C_d
10	4.70	0.250
15	5.25	0.275
20	5.85	0.300

Table 3: Estimated projected area and drag coefficient as functions of pitch angle

In steady cruise flight, the horizontal thrust component balances the aerodynamic drag force,

$$mg\tan(\theta) = \frac{1}{2}\rho S C_d v_{\text{cruise}}^2. \tag{5}$$

Solving for the cruise velocity yields

$$v_{\text{cruise}} = \sqrt{\frac{2mg\tan(\theta)}{\rho S C_d}}. \tag{6}$$

By iterating over different pitch angles, different cruise speeds can be calculated. Since the eVTOL is designed for human transport, the pitch angle is limited to a maximum of 20° for passenger comfort. Pitch angles of 10°, 15°, and 20° were therefore selected for analysis.

5.2 Momentum Theory

Momentum theory was implemented at an early stage of the modeling process to obtain a first-order estimate of the power consumption of the motors. The theory models the propeller as an ideal actuator disc, represented by an infinitesimally thin rotor that creates a pressure jump in the flow without adding rotational components such as vortices or turbulence in the airflow. The equations and theory are based on Snorri Gudmundsson [7], and the underlying assumptions are presented in Appendix B.

In hover, the flow is assumed to be axisymmetric and steady. The induced velocity at the actuator disc is denoted by w [m/s], while the far downstream velocity in the slipstream becomes

$$V_\infty = 2w. \tag{7}$$

The mass flow rate through the actuator disc is given by

$$\dot{m} = \rho A_p (V_{\text{disc}}) = \rho A_p w, \tag{8}$$

where A_p is the propeller disc area. Applying conservation of momentum, the thrust can be expressed as

$$T = \dot{m}(V_\infty - 0) = \rho A_p w(2w) = 2\rho A_p w^2. \tag{9}$$

For forward flight, the flow through the actuator disc is no longer zero upstream. Instead, the freestream velocity has a component normal to the rotor disc. The effective inflow velocity at the disc is therefore given by the sum of the induced velocity and the normal component of the freestream velocity,

$$V_\perp = V_\infty \sin(\theta), \tag{10}$$

assuming the rotor is tilted by an angle θ from the vertical.
The mass flow rate through the disc then becomes

$$\dot{m} = \rho A_p (V_\perp + w), \tag{11}$$

and applying conservation of momentum, the thrust can be expressed as

$$T = \dot{m}(2w) = 2\rho A_p w (V_\perp + w). \tag{12}$$

The induced power is then obtained as the product of thrust and induced velocity,

$$P = Tw. \tag{13}$$

By substituting the expression for thrust, the induced power can also be written as

$$P = \frac{T^{3/2}}{\sqrt{2\rho A_p}}. \tag{14}$$

5.3 Blade Element Momentum Theory

To obtain a more rigorous estimate of rotor thrust, Blade Element Theory (BET) is employed. Each rotor blade is discretized into small radial elements which are treated as two-dimensional airfoil sections. The sectional aerodynamic forces are computed using lift and drag coefficients, and the total thrust is obtained by integrating these contributions along the blade span.

The present eVTOL uses fixed rotors and flies in forward flight with an angled disc. In this case, the inflow is *skewed* relative to the rotor axis and varies with azimuth angle ψ, since the advancing and retreating sides has varied airspeeds, which requires a forward-flight rotor formulation. Therefore, Gudmundsson's axial-flight expressions [7] are combined with Bramwell's forward-flight kinematics [9].

The differential lift and drag acting on a blade element of width dr are

$$dL(r,\psi) = \frac{1}{2}\rho V_E(r,\psi)^2 \, c(r) \, C_l(\alpha(r,\psi)) \, dr, \tag{15}$$

$$dD(r,\psi) = \frac{1}{2}\rho V_E(r,\psi)^2 \, c(r) \, C_d(\alpha(r,\psi)) \, dr. \tag{16}$$

Following Bramwell, the local relative wind at a blade section is decomposed into a component U_t tangential to the plane of no-feathering and a perpendicular component U_p. [9] For small flapping angles and neglecting spanwise flow, Bramwell gives the following equations:

$$U_t(r,\psi) = \Omega r + V \sin a_f \sin \psi, \tag{17}$$

The inflow velocity component normal to the rotor disk is assumed to be uniform across the disk. It is modeled as the sum of the freestream component normal to the disk and a disk-averaged induced velocity

$$U_p = V_\perp + v_{i0} = V\cos(\alpha_f) + v_{i0} \tag{18}$$

In the present steady forward-flight analysis, these expressions capture the key physical effect: the chordwise relative wind is increased on the advancing side ($0 < \psi < \pi$) and decreased on the retreating side ($\pi < \psi < 2\pi$) [9].

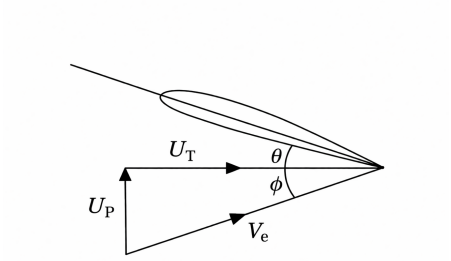


Figure 3: Illustration of Bramwells local wind speeds

The local inflow angle and the effective resultant velocity become

$$\phi(r,\psi) = \tan^{-1}\left(\frac{U_p(r,\psi)}{U_t(r,\psi)}\right), \tag{19}$$

$$V_E(r,\psi) = \sqrt{U_t(r,\psi)^2 + U_p(r,\psi)^2}. \tag{20}$$

The aerodynamic angle of attack of the blade element is then computed from the local pitch setting and inflow angle as the following equation. A representation of the angle can also be found in figure 4.

$$\alpha(r,\psi) = \beta(r) - \phi(r,\psi) - \alpha_{ZL}. \tag{21}$$

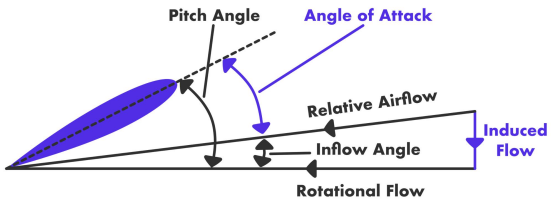


Figure 4: A representation of different angles [10]

Finally, the rotor thrust is obtained by projecting the sectional forces onto the rotor axis and averaging over azimuth,

$$T = N_b \frac{1}{2\pi}\int_0^{2\pi}\int_{R_{hub}}^{R} \left[\cos\phi(r,\psi)\, dL(r,\psi) - \sin\phi(r,\psi)\, dD(r,\psi)\right] dr\, d\psi. \tag{22}$$

To close the model, an induced-velocity estimate is required. In forward flight, Bramwell discusses Glauert's mean induced velocity relation [9]

$$T = 2\rho A v_{i0}(V_\perp + v_{i0}) \tag{23}$$

where $A = \pi R^2$ is the rotor disk area and V_R is the representative resultant velocity at the rotor. In this work, v_{i0} is solved iteratively together with Eq. (22).

Once the iterative solution for the induced velocity has converged, the aerodynamic forces acting on the rotor blades are fully determined, and the corresponding power requirement can be evaluated.

The sectional lift and drag forces are first resolved into axial and tangential components relative to the rotor plane. The tangential component gives rise to a torque contribution from each blade element, which can be expressed as

$$dQ = \frac{1}{2}\rho V_E^2\, c(r)\, c_t(r, \psi)\, r\, dr, \tag{24}$$

where c_t is the tangential force coefficient and r is the radial position of the blade element.

The total torque is obtained by integrating over the blade span and averaging over the azimuth angle, followed by multiplication with the number of blades. The required shaft power is then computed as

$$P = \Omega Q, \tag{25}$$

where Ω is the angular velocity of the rotor.

To enable the application of the BEMT formulation, several simplifying assumptions were required. In particular, detailed geometric data for commercially available propellers, such as chord and twist distributions, are often proprietary and not publicly accessible. An email was sent to T-Motor, the proposed manufacturer of the rotor blades on the eVTOL. The response can be seen in Appendix D where the request of data was not able to be provided.

To address this limitation, an existing BEMT implementation (Uploaded to github through the user "Cotri" [11]) was utilized, which included geometric data for a reference propeller (APC 10x7 Thin Electric). This dataset was adopted as a baseline, and the chord and radial dimensions were scaled proportionally to match the size requirements of the present eVTOL configuration. The propeller dimension for this eVTOL was approximated to a diameter of 1.5 meters, with inspiration taken from ehangs EHang184 [12]. The EHang184 eVTOL is quite similar in size and has approximately the same payload capabilities as the one in this report.

Furthermore, the original implementation by "Cotri" was developed for conventional propeller analysis and did not account for the specific operating conditions of an eVTOL in forward flight with a tilted rotor. Therefore, modifications were introduced to incorporate the effects of rotor disk incidence and azimuthal variation in the inflow, ensuring that the model better represents the intended flight regime.

An assumption made in this work is that tip losses are negligible. Tip losses arise due to the pressure equalization between the pressure and suction side at the blade tip, which generates tip vortices and reduces the effective lift and thrust produced by the outer section of the blade. To account for this effect, a tip loss factor can be introduced, commonly calculated using Prandtl's tip loss model. An alternative method to mitigate tip losses is the use of ducted fans (shrouded propellers). The presence of a duct reduces the strength of tip vortices and can improve the overall propulsive efficiency. However, tip losses are not entirely eliminated, but rather significantly reduced. With well-designed ducts and tight tip clearances, an increase in thrust compared to

an open propeller can be achieved [7]. The final CAD model, seen in figure 1d, includes ducts, primarily for safety reasons, but also with the added benefit of reducing tip losses.

6 Energy Consumption

6.1 Motor

When selecting suitable electric motors for an eVTOL application, the primary dimensioning parameter is the required power output to sustain flight. While several motor architectures are available, key performance indicators such as power-to-weight ratio and efficiency are of critical importance in aerospace applications.

The motors must be capable of continuously delivering the power levels estimated using the Blade Element Momentum Theory (BEMT), as defined in Eq 25. In addition to steady-state operation, the system should include a performance margin to accommodate transient power demands. These may arise from disturbances such as turbulence, maneuvering requirements, or other unforeseen flight conditions.

A relevant candidate for motor supply is Evolito [13], a company specializing in high-performance electric propulsion systems for aerospace applications. Their motors typically operate within a power range of 200–350 kW and are developed in collaboration with Vertical Aerospace, indicating applicability in similar aircraft configurations [14].

The mass of Evolito's motors, approximately 8–13 kg, aligns well with the weight constraints of this eVTOL platform. However, it should be noted that this figure represents the motor alone. The total propulsion system mass will increase when accounting for additional components such as inverters, power electronics, cabling, and thermal management systems.

6.2 Battery analysis

The battery system constitutes a central component of the eVTOL aircraft. Since both the propulsion motors and the low-voltage onboard systems depend entirely on electrical energy stored in the batteries, a thorough analysis of the battery requirements is necessary.

The aircraft is designed for an operational range of 30 km. The power required from the motors depends on the flight speed as well as the flight phase. The flight profile is divided into three distinct phases: take-off, cruise, and landing. Among these, the cruise phase is the most energy-consuming, as the motors must simultaneously provide lift to counteract gravity and thrust for forward motion.

To estimate the total power consumption, the power requirements for each flight phase must be evaluated. The power required by each motor can be expressed as a function of the thrust demand in the corresponding flight condition. During hover, the total thrust required equals the gravitational force acting on the aircraft. Assuming an ideal and symmetric configuration, each of the four motors is therefore assumed to provide one quarter of the total required thrust.

$$T_{motor} = \frac{mg}{4} \tag{26}$$

Deviations due to centre-of-mass offsets relative to the centre of volume are neglected for simplicity.

The static power required by each motor in hover is derived from momentum theory as presented by Snorri Gudmundsson. [7] The static induced power is given by Eq 14.

The thrust magnitude depends on the pitch angle θ according to Eq 2 as previously established. While the thrust required during take-off must exceed the gravitational force, the

corresponding increase in power demand is assumed to be approximately balanced by the reduced thrust requirement during landing. Consequently, these effects are neglected in the energy analysis, and the static hover power is used as a representative value for vertical flight segments.

During the cruise phase, steady flight requires equilibrium of both vertical and horizontal forces. The total thrust generated by the rotors must therefore satisfy two conditions. The horizontal component of the thrust must balance the aerodynamic drag, while the vertical component must balance the weight of the aircraft.

These conditions form a coupled system of equations, where the thrust is obtained from the BEMT formulation and depends on the flight speed, rotor pitch angle, and rotational speed. Due to the nonlinear nature of this system, an analytical solution is not readily available.

Instead, an iterative approach is employed. For a given rotor configuration, the thrust is evaluated over a range of flight speeds and operating conditions. The resulting thrust curves are then compared to the required force balance in order to determine whether a feasible equilibrium point exists, as shown in figure 6.

In addition, parametric variations in rotor speed and pitch angle are investigated to assess their influence on the achievable flight conditions. This allows identification of operating points where both vertical and horizontal force equilibrium can be satisfied.

The design mission requires a range of 30 km, which directly determines the required battery capacity. The energy required to complete this mission depends on the cruise power and, consequently, on the selected pitch angle.

The cruise time is determined from the kinematic relation

$$t = \frac{s}{v_{\text{cruise}}}, \tag{27}$$

where $s = 30$ km is the mission distance and v_{cruise} is the steady-state cruise speed corresponding to the given pitch angle. The cruise energy is then obtained as

$$E_{\text{cruise}} = P_{\text{cruise}}\, t \tag{28}$$

where P_{cruise} denotes the total propulsion power required in cruise flight. For a total aircraft mass of 650 kg, the resulting energy consumption for the considered pitch angles is presented in the results.

The vertical segments are assumed to consist of 30-60 seconds of take-off and 30-60 seconds of landing. Since these phases are performed in near-hover conditions, the energy consumption is assumed to be independent of pitch angle. The total mission energy is therefore obtained as the sum of cruise energy and vertical flight energy.

When determining the required battery capacity, several additional factors must be considered beyond the nominal mission energy. It would be imprudent to size the battery solely based on the calculated mission requirement without incorporating a safety margin.

Operational conditions in Singapore are characterized by high humidity, strong and rapidly changing winds, and frequent heavy rainfall. Adverse weather conditions may increase aerodynamic drag, require higher thrust levels, and consequently increase power consumption. Furthermore, additional energy may be required to compensate for system inefficiencies, battery aging, temperature effects, and potential deviations from the idealised flight model.

To account for these uncertainties, a safety factor is introduced in the battery sizing process. Let E_{mission} denote the total calculated mission energy. The required battery capacity is then defined as

$$E_{\text{battery}} = \gamma\, E_{\text{mission}},$$

where γ is the safety factor.

In this study, a conservative safety factor of $\gamma = 1.3$ is adopted. This corresponds to a 30% energy margin.

7 Noise pollution

To estimate the acoustic impact of the proposed multi-rotor eVTOL configuration, a semi-empirical propeller noise model is used. The approach follows the methodology presented in Gudmundsson [7] and provides a first estimate of the sound pressure level generated by propellers or rotors in farfield conditions.

The model is based on the idea that the overall sound pressure level (SPL) generated by a propeller can be expressed as the sum of several contributing factors. These contributions account for rotor tip speed effects, rotor geometry, propagation distance, directional effects and the number of propellers operating simultaneously.

The procedure therefore consists of computing a set of intermediate quantities, which are then combined to estimate the total sound pressure level experienced by an observer.

It should be noted that the original formulation of the model is primarily intended for conventional aircraft propellers operating under steady conditions. As such, it does not fully capture the acoustic characteristics of multi-rotor eVTOL configurations, where multiple interacting rotors and unsteady operating conditions introduce additional noise features such as complex tonal content.

What this model also does not account for is the urban environment in which the eVTOL will operate, neither how the generated noise is perceived by humans. Therefore, theory based on Nicholls [15] is considered to complement the SPL-based approach.

In particular, UAV noise is characterised by pronounced tonal components, high-frequency content, and temporal fluctuations arising from rotor dynamics. These features are not captured by a simple broadband sound pressure level model, but have been shown to significantly influence perceived annoyance. Furthermore, the surrounding urban soundscape plays an important role, where background noise levels and environmental context affect how intrusive the noise is perceived.

7.0.1 Rotor tip speed and tip Mach number

The acoustic emission from a propeller is strongly influenced by the blade tip velocity. Higher tip speeds increase compressibility effects and typically lead to higher noise levels. The tip velocity is computed as

$$V_{tip} = \sqrt{V_\infty^2 + \left(\frac{\pi\, RPM\, D_p}{60 \cdot 2}\right)^2} \tag{29}$$

The corresponding tip Mach number is given by

$$M_{tip} = \frac{V_{tip}}{a} \tag{30}$$

where the speed of sound is approximated as

$$a = \sqrt{\gamma RT} \tag{31}$$

7.0.2 Acoustic source strength

The acoustic source strength depends on both tip Mach number and shaft power:

$$FL_1 = 17.237 + 36.886 M_{tip} + 6.88 \ln(P_{HP}) \tag{32}$$

7.0.3 Rotor geometry contribution

The effect of rotor geometry is accounted for through

$$FL_2 = 32.551 - 8.312\ln(N_B) - 8.642\ln(D_p) \tag{33}$$

7.0.4 Propagation distance correction

The attenuation due to propagation distance is given by

$$FL_3 = \begin{cases} 16.29 - 0.5\ln(x_{ft}) & x_{ft} < 100 \\ 56 - 9.1202\ln(x_{ft}) & x_{ft} \geq 100 \end{cases} \tag{34}$$

The case for $x_{ft} \geq 100$ will primarily be used

7.0.5 Directional radiation effects

Directional effects are accounted for using a polynomial approximation:

$$\begin{aligned} DI = {} & 4.4464 \cdot 10^{-11}\theta^6 - 2.1049 \cdot 10^{-8}\theta^5 + 3.56 \cdot 10^{-6}\theta^4 \\ & - 0.0002575\theta^3 + 0.006073\theta^2 + 0.15363\theta - 8.1226 \end{aligned} \tag{35}$$

7.0.6 Multiple rotor correction

The contribution of multiple rotors is included through a discrete correction:

$$NC = \begin{cases} 0 & 1 \text{ rotor} \\ 3 & 2 \text{ rotors} \\ 5 & 3 \text{ rotors} \\ 6 & 4 \text{ rotors} \end{cases} \tag{36}$$

The proposed eVTOL configuration is using four rotors and thus $NC = 6$ will be used

7.0.7 Total sound pressure level

The total sound pressure level is obtained by summing all contributions:

$$SPL = FL_1 + FL_2 + FL_3 + DI + NC \tag{37}$$

7.0.8 Perceived noise level

To account for human perception, a perceived noise level correction is introduced. Auxiliary parameters are defined as

$$s = 0.2D_p - 1 \tag{38}$$

$$t = 2.25 - 2.5M_{tip} \tag{39}$$

The perceived noise level is then given by

$$PNL = SPL + PNL_{correction} \tag{40}$$

The $PNL_{correction}$ factor is approximated based on how many blades each rotor has. For a three bladed rotor this correction factor is given by

$$PNL_{correction} = PNL_{\text{3-blades}} = 7.168 + 2.511s + 3.402t - 1.720st \ \ [\text{dB}] \tag{41}$$

An approximate A-weighted level is obtained as

$$PNL_A \approx PNL - 14 \tag{42}$$

[7]

7.1 eVTOL in urban environment

The perception of noise from eVTOL aircraft differs from that of conventional aircraft due to differences in propulsion and operation. Conventional aircraft noise is typically dominated by broadband components with relatively smooth temporal variation during flyover. In contrast, multi-rotor eVTOL systems generate noise with stronger tonal components and more rapid fluctuations. This is mainly caused by the interaction of multiple rotors and variations in rotational speed during different flight phases. As a result, eVTOL noise can be perceived as more complex and more intrusive to humans, even at similar sound pressure levels. These differences are shown below where the frequency spectrum of eVTOL noise illustrates distinct tonal peaks compared to the smoother spectrum of conventional aircraft noise.

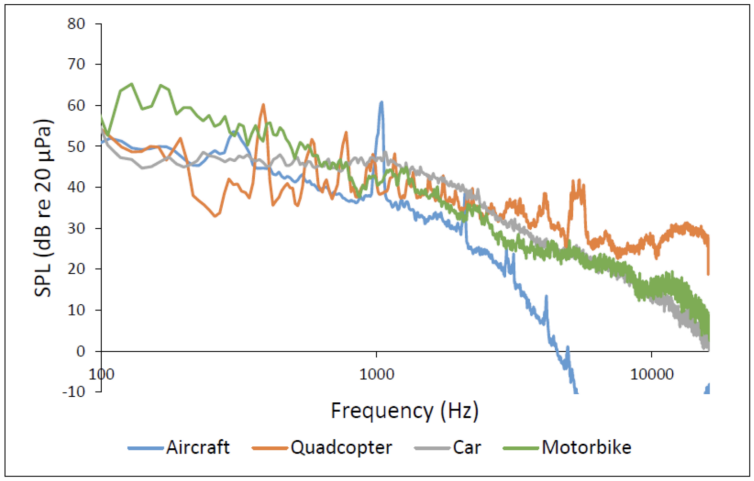


Figure 5: Frequency spectrum comparison between vehicles, adapted from Nichols

Additionally, eVTOL aircraft operate closer to the ground and in urban environments, where background noise and reflections from buildings influence how the sound is perceived. This means that the same sound level may result in different levels of annoyance depending on the surrounding environment.

To account for these effects, a modified perception-based indicator is introduced to complement the baseline noise model. The indicator is defined as

$$I_{eVTOL} = PNL_A + \Delta_{\text{tone}} + \Delta_{\text{HF}} + \Delta_{\text{fluct}} - \Delta_{\text{mask}} \tag{43}$$

where PNL_A represents the baseline A-weighted perceived noise level. The additional terms account for key characteristics of eVTOL noise that are not captured by conventional models. The tonal contribution Δ_{tone} reflects the presence of distinct frequency peaks associated with rotor dynamics, as illustrated in Figure 5. The high-frequency term Δ_{HF} represents the increased energy at higher frequencies, often perceived as sharper and more intrusive. Temporal variations in the noise signal are captured by Δ_{fluct}, arising from rotor interactions and unsteady operation. Finally, the masking term Δ_{mask} accounts for the influence of the surrounding urban soundscape, which can reduce the perceived prominence of the noise. I_{eVTOL} is intended as a comparative tool to highlight how eVTOL noise characteristics may differ from conventional aircraft predictions, rather than as an absolute measure of perceived annoyance. [15]

8 Regulations

Noise regulations will play a critical role in the market adoption and operational feasibility of eVTOL services in Singapore, particularly because vertiports are expected to be located in densely populated urban areas close to business districts, transport hubs, and residential zones. Singapore enforces strict environmental noise limits, with maximum permissible noise levels as low as 65 dBA during daytime and 55 dBA during nighttime near residential premises, reflecting the country's strong emphasis on minimizing noise disturbance in its high-density environment [16]. These limits will directly influence vertiport placement, flight paths, and operating hours, as excessive noise could lead to regulatory restrictions or public opposition. While these constraints present operational challenges, they also create a favorable environment for eVTOL adoption, as electric propulsion systems have the possibility to be quieter than traditional rotorcraft. [17] This reduced noise footprint enables eVTOL aircraft to operate more compatibly within dense urban environments, supporting their integration into Singapore's transport network while maintaining compliance with strict environmental standards.

Beyond environmental noise limits, the Air Navigation Act establishes the broader regulatory framework governing how eVTOL aircraft can be integrated into Singapore's airspace. All aircraft, operators, and aviation infrastructure must obtain the necessary certifications, licenses, and approvals before operation is permitted [18]. This includes airworthiness certification of the aircraft, operator certification, and approval of vertiports, which are regulated similarly to airports. These requirements represent a key barrier to entry but also provide a clear pathway for integration. From an engineering perspective, this makes certification driven design essential from the early development stages. Aircraft systems must demonstrate high levels of safety and reliability to meet regulatory standards and ensure approval for commercial operation in dense urban environments such as Singapore.

Several important factors influencing adoption can be directly addressed through engineering and operational decisions. Vertiport placement can be strategically planned in commercial districts, near transport hubs, or in less noise-sensitive areas to improve regulatory feasibility. Flight paths and operational procedures can also be optimized to avoid restricted airspace and integrate safely with existing air traffic.

However, there are also external constraints that cannot be directly controlled by manufacturers or operators. Airspace restrictions imposed by the Civil Aviation Authority of Singapore may limit available flight corridors, particularly near airports, military zones, or other protected areas [18]. Certification timelines, regulatory approval processes, and infrastructure licensing requirements are also determined by regulatory authorities and may introduce delays. While these factors cannot be controlled directly, they can be mitigated by aligning aircraft design with existing aviation standards and engaging with regulators early in the development process.

9 Economic possibilities

9.1 Background

For an eVTOL business to become profitable in Singapore, several key parameters must align. The main challenge is not necessarily technological development, but rather whether the market conditions are strong enough to support eVTOL operations as a long-term commercial service. In other words, even if the technology is ready, the critical question is whether Singapore can become a market where eVTOL air taxi services can scale sustainably and generate consistent demand.

A central uncertainty is how consumers will respond to a new type of transportation provider. Would there be sufficient interest for eVTOL services to achieve long-term success, or would demand remain limited to a niche segment? If the goal is to build a profitable market, the focus must be on creating clear value for customers, primarily through time savings, convenience, and reliability, while also ensuring that safety, trust, and social acceptance develop over time.

Once demand is established, the next major factor is the economic cost structure. Even with strong customer interest, profitability will depend on fleet costs, operating expenses, maintenance requirements, and the full service and support system needed to keep aircraft flying at high utilization rates. In practice, the eVTOL business case depends heavily on unit economics: how much revenue each aircraft can generate per day compared to its total cost to own and operate.

In addition, eVTOL operations require large upfront investment in infrastructure such as vertiports, charging systems, maintenance facilities, and integration with airspace management systems. Because these investments are significant, it becomes important to understand the timeline for return on investment. How quickly could an operator realistically reach break-even, and what level of operational scale would be required before the business becomes financially sustainable?

Finally, public sector involvement may play a decisive role, especially in the early phase. If infrastructure costs are high relative to early demand, government support may be needed to reduce risk and make implementation realistic. This raises the question of what role subsidies, public-private partnerships, or regulatory fast-tracking could play in enabling eVTOL commercialization without requiring the operator to carry the entire financial burden upfront.

9.2 Global outlook

If local eVTOL operations are to become viable in a market such as Singapore, economic feasibility must first be demonstrated at a global scale. The capital intensity, certification complexity, and infrastructure requirements of multirotor eVTOL systems demand scalable business models that function across multiple cities. Without evidence of sustainable profitability in the global market, localized deployment risks remaining technologically feasible but economically unviable.

The global economic outlook for the eVTOL, particularly multirotor configurations, must be evaluated within the broader context of both the traditional mobility market and the emerging "low-altitude economy." The global mobility market today is valued at approximately $ 7,800 billion, with road mobility accounting for more than 80% of total value. Against this backdrop, the projected base-case market potential for passenger eVTOL services in 2035 is approximately $32 billion, of which $21 billion is expected to stem from intracity air taxi services. Vertical mobility will therefore remain a niche within global transportation, but one with strategic relevance in high-density urban corridors. [19]

According to Bank of America Global Research, the global civil adoption of eVTOL aircraft is expected to grow at a compound annual growth rate of 62% during 2025–2030. Furthermore, global civil eVTOL units in operation are projected to reach approximately 77,000 units by

2035. These projections indicate significantly accelerated adoption compared to earlier industry assumptions and suggest that scaling dynamics may materialize more rapidly than initially expected. [20]

Despite promising growth forecasts, profitability remains contingent upon large-scale deployment and ecosystem coordination. The base-case scenario requires more than 30 cities globally to establish intracity air taxi operations by 2035. Achieving economic relevance necessitates a global network of more than 15,000 eVTOL aircraft operating from approximately 1,000 to 2,500 vertiports. Without this level of geographic reach and infrastructure deployment, the high fixed costs associated with development and certification cannot be spread across a sufficiently large number of operations. As a result, unit costs remain elevated and profitability becomes difficult to achieve. Examining the global scale potential is therefore essential in order to determine whether the industry can reach the volume necessary to support sustainable economics. Only if such scale is realistic worldwide can local markets, such as Singapore, expect long-term commercial viability. [19]

Bank of America Global Research further projects that the broader urban air mobility market could reach approximately $23 billion by 2035, with longer-term expansion toward $61 billion and up to $210 billion by 2045 under extended growth scenarios. Civil eVTOL units in operation are forecast to increase significantly over the coming decades, reflecting strong anticipated adoption momentum. At the same time, eVTOLs are positioned primarily as a supplement to existing transportation modes rather than as direct replacements. In particular, for travel distances between 150–400 km, helicopters and eVTOLs demonstrate competitive travel times relative to ground transport and conventional aviation. For shorter urban corridors, eVTOL services are expected to occupy a premium segment, characterized by higher fares but offering time savings, increased privacy, and differentiated travel experience. [20]

9.3 Demand and market adoption in Singapore

Market demand for multirotor eVTOL services in Singapore must be evaluated in the context of urban congestion and travel efficiency. According to the TomTom Traffic Index (2025), Singapore reports an average congestion level of 43.9%, with approximately 81 hours lost annually during rush hour. Morning congestion reaches 60.6%, while evening congestion peaks at 69.6%, with average speeds dropping to 30.2 km/h and 21.2 km/h respectively during peak periods. These figures indicate that even in a highly optimized and well managed transport system, peak-hour delays remain structurally embedded in daily mobility patterns. For time sensitive commuters and premium travelers, persistent congestion creates a potential willingness to pay for faster, more reliable alternatives. [21]

However, market adoption will depend not only on congestion levels but also on the relative performance advantage and regulatory feasibility of eVTOL services. Airport transfers represent a potentially strong demand segment due to high time sensitivity, yet implementation is more complex because of airspace restrictions, coordination with Changi Airport operations, and cross agency regulatory approvals. Cross border travel presents another significant demand driver. Ferry connections between Singapore and Batam are among the busiest short-distance international sea routes in Southeast Asia, with several million passengers transported annually prior to the pandemic, indicating structurally strong leisure and commuter demand. The distance from central Singapore to Batam is approximately 20 km, which at an assumed eVTOL cruising speed of 130 km/h would imply a flight time of roughly 9–10 minutes excluding terminal procedures. Such travel times would significantly undercut ferry durations whos duration is around an hour between the countries. If regulatory frameworks permit cross-border low-altitude operations, the Singapore–Batam corridor could represent a high value early adoption case for

multirotor eVTOL deployment. [22]

Singapore's position at the very high end of the global economic spectrum further strengthens the potential demand for premium transportation services such as eVTOL. Singapore has one of the highest GDP per capita levels in the world, reflecting exceptionally high average productivity and income. The country is also home to a large concentration of millionaires and billionaires [23], supported by its favorable tax environment, which includes no capital gains tax and relatively low personal income tax rates compared to other major financial centers. [24] This tax efficiency attracts high-net-worth individuals, business executives, and international investors. These individuals typically have significantly higher willingness to pay for convenience, time savings, and premium services, and are less sensitive to transportation costs relative to their income. As a result, the presence of a substantial ultra-high-income population increases the serviceable obtainable market for eVTOL services, particularly during early deployment phases when costs are expected to remain high. This high-income segment is likely to serve as the primary early adopter group, enabling initial commercial viability and supporting gradual expansion toward broader market segments as the technology matures and costs decrease.

In addition to its large population of ultra-high-net-worth individuals, Singapore's strong middle- and upper-middle-income segments further support the long-term adoption potential of eVTOL services. The median gross monthly income of full-time employed residents reached SGD 5,775 (around USD 4,300) in 2025, with sustained income growth over the past decade [25]reflecting increasing purchasing power across a broad portion of the population. While initial eVTOL adoption is expected to be concentrated among high-income early adopters, the relatively high baseline income level of the median population indicates that a substantial share of residents may become economically capable of using such services as costs decrease. As technological advancements, operational scaling, and infrastructure development reduce per-passenger costs, eVTOL services are expected to transition from an exclusive premium offering toward a more widely accessible transport option. This gradual expansion into the upper-middle and middle-income segments will significantly increase the serviceable available market and support long-term demand growth beyond the initial high-income user base.

9.4 Cost structure and unit economics

In an economic analysis of eVTOL operations, it is necessary to distinguish between fixed and variable costs. Fixed costs represent the baseline investment required to enable operations, independent of flight activity, while variable costs are directly linked to the number of flights and operational usage.

The largest fixed cost is associated with infrastructure, including investments in vertiports, charging systems, and maintenance facilities required to support operations. In addition, significant costs arise from the acquisition of the eVTOL aircraft itself, including propulsion systems, batteries, and structural components. Certification and regulatory approval also contribute substantially, as extensive testing and compliance with aviation standards are required before commercial operation can begin. Although these costs are independent of the number of flights, their impact per flight decreases with higher utilization, making operational efficiency a key factor.

Variable costs are directly associated with the operation of the eVTOL and therefore scale with the number of flights performed. A primary component is the energy consumption required for each mission, which depends on flight duration and power demand. In addition, maintenance costs contribute significantly, including wear on components such as batteries, propellers, and motors, as well as periodic replacements. Operational expenses such as ground handling, personnel, and insurance may also be considered variable to some extent. [26]

9.5 Achieve Profitability

There are multiple paths to achieving profitability, depending on how resources are allocated and operational strategies are defined. For instance, an operator may choose to invest in a more premium eVTOL experience in order to justify higher ticket prices, or instead focus on minimizing costs to offer a more competitive service. Operational parameters such as flight speed and altitude may also influence both cost and revenue. In addition, factors such as partnerships, advertising, marketing strategies, how many eVTOLs are in operation and potential subsidies will play an important role in shaping the overall business case.

Given the number of variables involved, it is not possible to provide a single definitive answer to whether eVTOL operations will be profitable in Singapore. However, in Section 9.5.1, a proposed scenario is presented to illustrate how profitability could potentially be achieved under a set of reasonable assumptions. The analysis will therefore focus on estimating the time required to reach break-even. Assumptions are chosen based on own calculations as well as reasonable values compared to existing current eVTOL operations. [12] [27–30]

9.5.1 Economic Analysis

Using the basic formulas from Appendix E along with the assumptions in Section 3 an estimate of time required to reach enough revenue to counter all the pre operational investments is calculated. Additionally, the model will output profit per flight as well as profit margin which is valuable when trying to optimize profit.

10 Result

10.1 Momentum Theory and BEMT

The AOA was analyzed in the range of 10° to 25° with increments of 5°. The calculated thrust over different airspeed for different angles of attack are presented below, for the motor RPM of 3000.

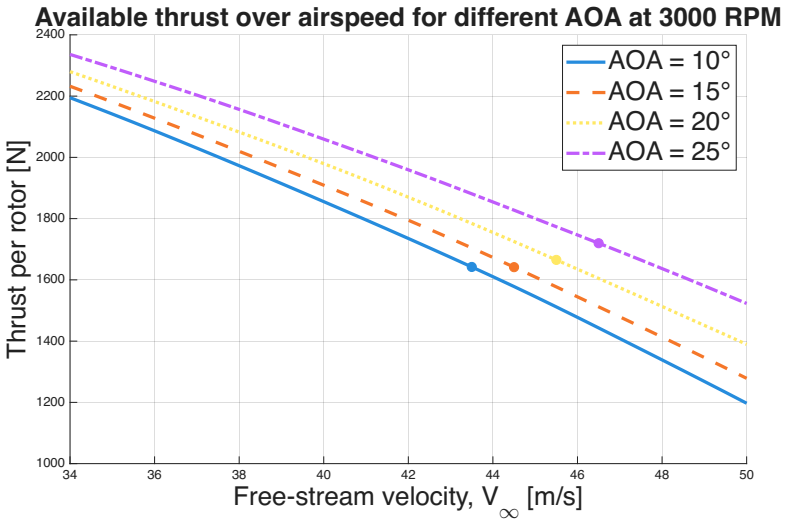


Figure 6: Available thrust over airspeed for 3100 RPM, with closest equilibrium points marked

The required RPM to achieve equilibrium, the range reflects variations due to pitch angle.

Number of Blades	Required RPM Range
2	3150 – 3350
3	2950 – 3050
4	2700 – 2900

Table 4: Required rotor RPM for equilibrium as a function of blade count.

The calculated cruise velocity is ranging from 140-170 km/h, also depending on pitch angle.

The total energy needed for take-off and landing is around 4.5 kWh. The range of required power is between 80-96 kW, leading to energy values shown in figure 5:

Pitch Angle (deg)	Cruise Speed (km/h)	Cruise (kWh)	T/O & L (kWh)	Total (kWh)
10	144	66.66	9	75.6
15	158	66.11	9	75.1
20	167	64.7	9	73.7
25	172	64.5	9	73.5
30	180	64.5	9	73.5

Table 5: Energy breakdown and total energy required as a function of pitch angle

The battery capacity must exceed 98.3 kWh after safety factor implementation.

10.2 Noise Pollution

Table 6 is a representation of the altitudes that the eVTOL is allowed to fly during daytime and nighttime, respectively, in accordance with Singaporean regulations. It should be remembered that 65dBA is the limit for daytime and 55dBA is the limit for nighttime.

Pitch Angle (deg)	Cruise Speed (km/h)	Cruise Altitude (m)	PNL_A (dBA)
0	0	10	89
10	144	540	65
10	144	1690	55
15	158	610	65
15	158	1940	55
20	167	690	65
20	167	2200	55

Table 6: Cruise altitude and perceived noise level as functions of pitch angle

10.3 Economic Analysis

Profit per flight is 108USD and with revenue per flight of 430USD the profit margin is 25.22%. With an initial total investment of 44 million dollars the payback time is 20.29 years.

11 Discussion

Results obtained from the BEMT model and the subsequent energy analysis provide a good estimate of the aerodynamic and energetic feasibility of the proposed eVTOL configuration.

However, several assumptions and modeling limitations must be critically examined in order to assess the reliability of these results and their implications for real-world operation.

The BEMT results indicate that sufficient thrust can be generated to meet both vertical and horizontal force requirement within the studied operating range. However, it is important to emphasize that the model provides available thrust for a given rotor configuration, rather than the actual thrust produced during steady flight. The actual operating point is instead determined by the intersection between the required force balance and the available thrust envelope. This means that the feasibility of a given flight condition depends on whether the required thrust lies below the predicted thrust curve. The most reliable method for determining the cruise speed corresponding to different pitch angles would be experimental validation using a physical prototype. This would allow direct measurement of aerodynamic performance under realistic operating conditions.

However, existing eVTOL concepts typically operate at cruise speeds in the range of 120–150 km/h, which suggests that the values obtained in this study are within a realistic and plausible range.

Despite providing a more detailed representation than momentum theory, the BEMT model still relies on several simplifying assumptions. In particular, blade–blade aerodynamic interference has not been accounted for. This effect can become significant in multi-bladed rotors, where wake interactions may reduce overall efficiency and alter the effective angle of attack distribution. A possible way to mitigate blade–blade interference is to reduce the number of blades per rotor. This however introduces a trade-off, as fewer blades require higher rotational speeds to generate the same thrust.

Furthermore, the use of scaled propeller geometry from an existing dataset introduces uncertainty, as real eVTOL propellers are likely optimized for different Reynolds numbers and operating conditions.

The energy analysis highlights that the cruise phase dominates the total mission energy consumption, which is consistent with the fact that both lift and forward thrust must be maintained simultaneously during this phase, and that the main mission for an eVTOL is to transport passengers over a distance.

However, the battery sizing is based on an idealized mission profile and assumes constant efficiency and steady operating conditions. In practice, factors such as battery degradation, temperature sensitivity, and power electronics losses will reduce the usable energy capacity. Additionally, environmental conditions such as high humidity, temperature, and wind variability in Singapore may further increase energy demand, reinforcing the need for conservative safety margins.

11.1 Noise Pollution

Results show that the maximum noise level experienced by ground staff is approximately 89 dBA. This appears reasonable, as personnel in close proximity to the aircraft would likely use hearing protection such as earmuffs. The same applies to passengers, who are expected to be among the most exposed to noise during operation.

When analyzing the allowed cruise altitudes, a clear relationship can be observed between pitch angle, flight speed, and noise constraints. Lower pitch angles result in lower flight speeds, which in turn allow the eVTOL to operate at lower altitudes while still complying with noise regulations. For daytime operations, a cruising altitude of approximately 540 meters satisfies the 65 dBA limit. This is a reasonable altitude, as it is above all buildings in Singapore while still allowing efficient take-off and landing without excessive time or energy consumption. For nighttime operations, where the noise limit is stricter at 55 dBA, the required cruising altitude

increases significantly to around 1690 meters. This introduces additional challenges, as climbing to such altitudes reduces both time efficiency and energy efficiency. As a result, it is unlikely that nighttime operations would be feasible without further technical improvements aimed at reducing noise emissions. However, certain routes may still be viable during nighttime. For example, flights between Singapore and Batam could potentially operate since most of the flight path would be over water, reducing the impact on populated areas. This route is also highly time-efficient and in high demand, which strengthens its feasibility. One possible solution to further reduce noise impact is to place vertiports near coastal areas or slightly offshore, minimizing disturbance for people living nearby.

An important limitation of the current analysis is that the effect of ducted rotors has not been explicitly included in the noise model. As discussed earlier in the report, the ducts reduce tip vortices and lead to smoother airflow, which is expected to lower noise emissions. Since this effect is not captured in the Gudmundsson model [7], the actual noise levels may be somewhat overestimated.

It is also important to note that the Gudmundsson model [7] is primarily developed for conventional aircraft propellers and does not fully represent the characteristics of multi-rotor eVTOL systems. Equation 43 addresses this by introducing additional correction terms. In particular, the masking term is likely to have a significant impact in an urban environment such as Singapore, where high building density and existing background noise can reduce the perceived annoyance of the eVTOL. During daytime, the high level of background noise in the city may help mask the eVTOL sound. However, during nighttime, when ambient noise levels are lower, the eVTOL noise will become more noticeable. Combined with tonal components and higher-frequency noise, this may negatively affect public perception.

Public acceptance will be a key factor for successful implementation, as the residents of Singapore represent the primary user base of the service. While noise remains a challenge, the fact that the eVTOL operates using fossil-free energy is a clear advantage, as it eliminates local emissions and contributes to a cleaner urban environment.

11.2 Economic Outlook

The economic model indicates a profit of approximately 108USD per flight, which appears reasonable given the assumptions made. However, the corresponding profit margin of 25% is relatively modest. Considering the significant contribution of variable costs such as electricity, ground operations, and maintenance there is limited room for substantial margin improvement without structural changes to the cost base. While increasing ticket prices may seem like a straightforward way to improve profitability, the situation is more complex. Ticket pricing must be treated as an optimization problem, where price levels are balanced against demand in order to maintain high load factors and maximize fleet utilization. Achieving this balance is critical, as operating partially filled flights would significantly reduce profitability. A more detailed analysis of demand fluctuation and future outlook would therefore be required to accurately assess the true profit potential of the service. It is also important to consider that demand is expected to increase over time. As market adoption grows, it is likely that pricing flexibility will improve, potentially allowing higher ticket prices in later stages of operation. This suggests that profitability may improve as the system matures.

In the current economic model, the fleet size is assumed to be 10 eVTOLs. While this may be a reasonable estimate for a medium term scenario, it is likely optimistic for early stage deployment, where fleet sizes may initially be limited to one or two aircraft. Conversely, in a more mature phase, the fleet could expand significantly, potentially exceeding 20 units. From this perspective, assuming an average fleet size of 10 over the first 20 years of operation may be justified, but the

model would benefit from explicitly incorporating fleet scaling over time.

The estimated payback time for this scenario is approximately 20 years, which is relatively long but not unreasonable given the high initial investment of approximately 44 million USD. Combined with a moderate profit margin, this naturally leads to an extended payback period. However, it is encouraging that the model indicates positive unit economics, meaning that each flight generates profit. This is a critical factor, as it suggests that the business model has the potential to become attractive if scalability can be achieved.

A key assumption underlying the feasibility of this service is the significant level of government support. In this model, the Singaporean state is assumed to subsidize 80% of infrastructure costs, particularly for vertiports. This corresponds to approximately 80 million USD, which is a very high level of public investment. While such support may seem ambitious, similar initiatives have been observed in early adopter regions such as Dubai [27], suggesting that comparable investments are not unrealistic. Nevertheless, it is uncertain whether the Singapore government would be willing to allocate such resources, particularly given that early adoption is likely to be limited to high-income users. From a policy perspective, this could raise concerns regarding accessibility and equitable use of public funds. An alternative scenario is that private investors contribute to infrastructure development in exchange for equity, revenue sharing, or long-term operational rights. Public–private partnerships may therefore represent a more realistic pathway for implementation.

Finally, it is important to emphasize that the economic viability of eVTOL operations is strongly dependent on scalability. Operating within a single market, such as Singapore, is unlikely to be sufficient to recover the substantial upfront investment. However, by expanding across multiple cities and countries, both demand and operational efficiency can increase, while fixed costs such as design and prototyping are distributed across a broader network. This scaling effect is likely to be a decisive factor in making the business case attractive in the long term.

12 Conclusion

The results of this study indicate that it is feasible to develop a functioning eVTOL aircraft configured as a quadcopter with fixed rotors. Provided that sufficiently powerful motors are employed, the generated thrust can exceed the vehicle's weight and enable sustained flight.

The estimated cruise speed is found to lie in the range of approximately 140–170 km/h, depending on the selected pitch angle. Furthermore, the energy analysis suggests that a battery capacity of around 100 kWh is sufficient to support vertical takeoff, a 30 km cruise segment, and landing, while maintaining a safety margin of approximately 30 %.

Nighttime operations in the city center are currently not feasible due to strict noise constraints. Meeting these requirements would necessitate flight at altitudes approaching 2000 m, which is neither energy efficient, time efficient, nor economically viable. However, an exception may be possible for routes such as the Singapore–Batam path, where flights occur primarily over water and may be subject to less restrictive noise considerations. In contrast, daytime operations appear feasible, with required cruising altitudes of approximately 600 m, which are operationally achievable.

The economic analysis indicates that eVTOL operations can achieve positive unit economics under the specific assumptions applied in this study. However, these results are highly sensitive to the chosen input parameters, including demand levels, fleet utilization, and cost estimates. The long payback period and substantial upfront investment further imply that overall profitability is not guaranteed.

13 Appendix

A Open VSP Cross sectional area

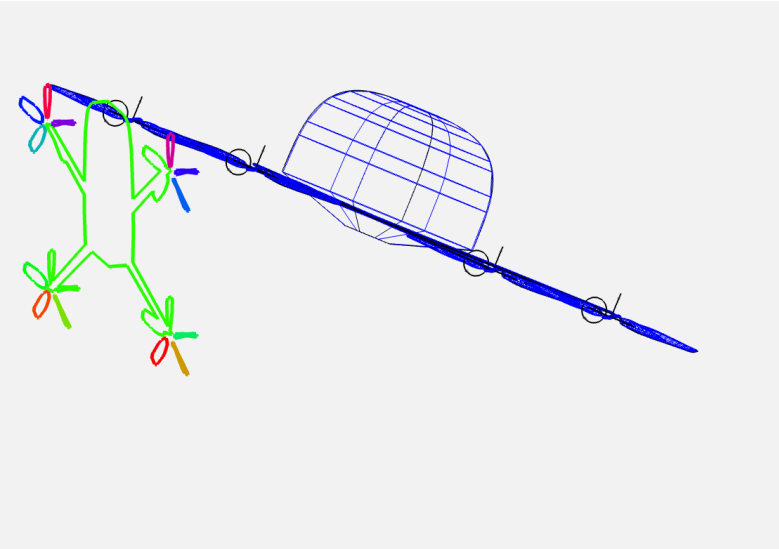


Figure 7: eVTOL illustration from OpenVSP

B Momentum Theory Assumptions

1. The propeller is replaced by an infinitesimally thin actuator disc that offers no resistance to the flow passing through it.
2. The actuator disc is uniformly loaded; hence, the flow through the disc is uniform and experiences uniform acceleration.
3. A control volume encloses the streamtube and clearly separates the flow passing through the disc from the surrounding air.
4. The flow outside the streamtube maintains constant stagnation pressure, implying that no work is imparted to it.
5. In the far field, the streamlines upstream and downstream of the actuator disc are parallel. Consequently, the static pressure within the streamtube far from the disc equals the ambient pressure.
6. The propeller does not impart any rotational (swirl) component to the flow.
7. The flow is assumed to be steady, inviscid, and incompressible.

C Drone model for approximation of Cd



Figure 8: UAS 1 drone from Hammers freefall analysis [8]

D Email from T-Power

Hi Rikard,

Hope this email find you well. This is Hugh Sun, if you have any questions or needs in the future, you can contact me directly.

I apologize for the late reply. After confirming with the engineering department, the data you requested is confidential internal information and cannot be provided directly to customers.

Thank you for your support and understanding of T-Motor.



Please double check with us via wechat, skype, phone call (number of which can be found on our website) or any other means of direct communication if you receive notification of bank account change.Please pay attention to property safety.

Best regards,

Hugh Sun

Sales & Tech Department

T-MOTOR®
THE SAFER PROPULSION SYSTEM

Figure 9: Recieved email from T-Motor regarding blade geometry data

E Economic model equations

The economic analysis is based on a simplified cost and profitability model for a fleet of eVTOL aircraft. The model separates variable costs, fixed costs, total operating cost, revenue, profit, and payback time.

E.1 Fleet utilization

The total number of annual flights in the fleet is defined as

$$N_{\text{annual}} = N_{\text{fleet}} \cdot N_{\text{annual,eVTOL}} \tag{44}$$

The total number of flights over the assumed economic lifetime is

$$N_{\text{lifetime}} = N_{\text{annual}} \cdot T_{\text{economic}} \tag{45}$$

E.2 Variable costs

The energy cost per flight is calculated as

$$C_{\text{energy}} = E_{\text{flight}} \cdot p_{\text{electricity}} \tag{46}$$

The total battery cost is

$$C_{\text{battery,total}} = E_{\text{battery}} \cdot p_{\text{battery}} \tag{47}$$

The battery cost allocated to each flight is then

$$C_{\text{battery,flight}} = \frac{C_{\text{battery,total}}}{N_{\text{cycles}}} \tag{48}$$

The total variable cost per flight is given by

$$C_{\text{variable}} = C_{\text{energy}} + C_{\text{battery,flight}} + C_{\text{maintenance}} + C_{\text{ground}} + C_{\text{insurance}} \tag{49}$$

E.3 Fixed costs

The total fleet acquisition cost is

$$C_{\text{fleet,total}} = N_{\text{fleet}} \cdot C_{\text{eVTOL}} \tag{50}$$

The flight time in hours is

$$t_{\text{flight}} = \frac{t_{\text{flight,min}}}{60} \tag{51}$$

The aircraft cost per flight is calculated from the hourly depreciation

$$C_{\text{aircraft,hour}} = \frac{C_{\text{fleet,total}}}{N_{\text{fleet}} \cdot T_{\text{eVTOL,hours}}} \tag{52}$$

$$C_{\text{aircraft,flight}} = C_{\text{aircraft,hour}} \cdot t_{\text{flight}} \tag{53}$$

The effective infrastructure cost after subsidy is

$$C_{\text{infrastructure,eff}} = C_{\text{infrastructure}} \left(1 - f_{\text{subsidy}}\right) \tag{54}$$

The infrastructure and certification costs allocated to each flight are

$$C_{\text{infrastructure,flight}} = \frac{C_{\text{infrastructure,eff}}}{N_{\text{lifetime}}} \tag{55}$$

$$C_{\text{certification,flight}} = \frac{C_{\text{certification}}}{N_{\text{lifetime}}} \tag{56}$$

The total fixed cost per flight is

$$C_{\text{fixed}} = C_{\text{aircraft,flight}} + C_{\text{infrastructure,flight}} + C_{\text{certification,flight}} \tag{57}$$

E.4 Total cost metrics

The total cost per flight is

$$C_{\text{total}} = C_{\text{fixed}} + C_{\text{variable}} \tag{58}$$

The cost per kilometer is defined as

$$C_{\text{km}} = \frac{C_{\text{total}}}{R} \tag{59}$$

The effective number of paying passengers is

$$N_{\text{pass,eff}} = N_{\text{pass}} \cdot f_{\text{load}} \tag{60}$$

The cost per passenger is then

$$C_{\text{passenger}} = \frac{C_{\text{total}}}{N_{\text{pass,eff}}} \tag{61}$$

E.5 Revenue and profit

The revenue per flight is calculated as

$$R_{\text{flight}} = p_{\text{ticket}} \cdot N_{\text{pass,eff}} + R_{\text{ads}} \tag{62}$$

The profit per flight after corporate tax is

$$\Pi_{\text{flight}} = (R_{\text{flight}} - C_{\text{total}})\, f_{\text{tax}} \tag{63}$$

The profit margin is given by

$$M_{\text{profit}} = \frac{\Pi_{\text{flight}}}{R_{\text{flight}}} \tag{64}$$

E.6 Payback time

The annual profit is

$$\Pi_{\text{annual}} = \Pi_{\text{flight}} \cdot N_{\text{annual}} \tag{65}$$

The total initial investment is defined as

$$C_{\text{initial}} = C_{\text{infrastructure,eff}} + C_{\text{certification}} + C_{\text{fleet,total}} \tag{66}$$

Finally, the payback time is estimated as

$$T_{\text{payback}} = \frac{C_{\text{initial}}}{\Pi_{\text{annual}}} \tag{67}$$

F Use of Artificial Intelligence Tools

Artificial intelligence tools were used as supporting aids during the preparation of this report. The tools were primarily utilized to assist with LaTeX formatting, generation and refinement of MATLAB plotting scripts, grammar and language review, and guidance regarding citation formatting and referencing practices.

References

[1] EHang. About ehang. [Online]. Available: https://www.ehang.com/about/

[2] IEEE Spectrum. Joby aviation's air taxi moves closer to reality. [Online]. Available: https://spectrum.ieee.org/joby-air-taxi

[3] HeliCentre, "Frequently asked questions." [Online]. Available: https://www.helicentre.eu/en/faq/

[4] R. A. McDonald and J. R. Gloudemans, "Open vehicle sketch pad: An open source parametric geometry and analysis tool for conceptual aircraft design," in *AIAA SCITECH 2022 Forum*. American Institute of Aeronautics and Astronautics, 2022.

[5] P. A. S. F. Silva, P. Tsoutsanis, J. R. P. Vaz, and M. M. Macias, "A comprehensive cfd investigation of tip vortex trajectory in shrouded wind turbines using compressible rans solver," *Energy*, 2024.

[6] EHang. Ehang 216s. [Online]. Available: https://www.ehang.com/ehang216s/

[7] Snorri Gudmundsson, *"General Aviation Aircraft Design, Applied Methods and Procedures"*, 2022 second edition.

[8] T. Hammer, J. Quitter, J. Mayntz, J.-M. Bauschat, P. Dahmann, F. Götten, S. Hille, and E. Stumpf, "Free fall drag estimation of small-scale multirotor unmanned aircraft systems using computational fluid dynamics and wind tunnel experiments," *CEAS Aeronautical Journal*, vol. 15, 2024.

[9] A. R. S. Bramwell, *Bramwell's Helicopter Dynamics*, 2000 Second edition.

[10] Jop Dingemans. (2024) Helicopter rotor blade aerodynamics: The basics step by step. [Online]. Available: https://pilotswhoaskwhy.com/2024/07/28/helicopter-rotor-blade-aerodynamics-the-basics-step-by-step/

[11] Cotri. Bemt. [Online]. Available: https://github.com/cotri/BEMT

[12] Airport Technology. (2016) Ehang 184 autonomous aerial vehicle (aav). [Online]. Available: https://www.airport-technology.com/projects/ehang-184-autonomous-aerial-vehicle-aav/

[13] Evolito. Axial flux motors. [Online]. Available: https://evolito.aero/axial-flux-motors/

[14] Evoltio. Vertical aerospace selects evolito as electric engine partner for valo evtol. [Online]. Available: https://evolito.aero/vertical-aerospace-selects-evolito-as-electric-engine-partner-for-valo-evtol/

[15] R. K. Nicholls, "An investigation into the impact of unmanned aerial vehicles on soundscape perception in urban environments," Master's thesis, University of Salford, 2021.

[16] National Environment Agency, Housing & Development Board, and Municipal Services Office, Singapore, "Infosheet on noise regulations and guidelines in singapore," 2023. [Online]. Available: https://www.nea.gov.sg/

[17] W. Raza *et al.*, "Noise prediction and mitigation for uas and evtol aircraft," *Drones*, vol. 9, no. 8, p. 577, 2025.

[18] Government of Singapore, "Air navigation act 1966 (2020 revised edition)," 2020.

[19] Porsche Consulting, "The economics of vertical mobility: A guide for investors, players, and lawmakers to succeed in urban air mobility," 2021.

[20] Bank of America Institute, "The "low-altitude" economy is taking off," June 2025.

[21] TomTom, "Tomtom traffic index 2025: Singapore traffic overview," 2025.

[22] Maritime and Port Authority of Singapore, "Port statistics 2019," 2019.

[23] Global Finance Magazine, "Richest countries in the world 2025," 2025. [Online]. Available: https://gfmag.com/data/richest-countries-in-the-world/

[24] PricewaterhouseCoopers (PwC), "Singapore – worldwide tax summaries overview," 2025. [Online]. Available: https://taxsummaries.pwc.com/singapore

[25] Ministry of Manpower, Singapore, "Summary table: Income," 2026. [Online]. Available: https://stats.mom.gov.sg/pages/income-summary-table.aspx

[26] K. Hill. (2025) Variable costs, fixed costs, total costs: How do they differ? Accessed: 2026-04-20. [Online]. Available: https://preferredcfo.com/insights/variable-costs-fixed-costs-total-costs-how-do-they-differ

[27] Ken Research. (2025, Nov.) Uae urban air mobility market report size, share, growth drivers, trends, opportunities & forecast 2025–2030. [Online]. Available: https://www.kenresearch.com/uae-urban-air-mobility-market

[28] C. Stonor. (2022, Nov.) evtol operational costs and profit: "does anyone have a clue... yet?". [Online]. Available: https://evtolinsights.com/evtol-operational-costs-and-profit-does-anyone-have-a-clue-yet/

[29] P. Butterworth-Hayes. (2024, Dec.) "maintenance will account for 37% of evtol airport shuttle costs" – new research. [Online]. Available: https://www.urbanairmobilitynews.com/air-taxis/maintenance-will-account-for-37-of-evtol-airport-shuttle-costs-new-research/

[30] Osome. (2025, Oct.) Singapore corporate income tax: Rates, system, reliefs. [Online]. Available: https://osome.com/sg/blog/singapore-corporate-income-tax-rates-system-reliefs/